

# MODULE IV

## SITE PLANNING AND BUILDING RULES

---

# INTRODUCTION

- ▮ Location of building has utmost importance in bringing a tranquil atmosphere in the dwelling.
- ▮ Properly located site with good foundation soil brings down the construction cost.

# FACTORS GOVERNING SELECTION OF SITE

- ▣ Site should be fairly level with good quality of soil.
- ▣ Should be well connected by roads and other means of transport.
- ▣ Should have good communication facilities.
- ▣ Electricity, water and sewer lines should be available.
- ▣ Should be away from hazardous industries.
- ▣ Location should be calm but reasonably developed.
- ▣ Should have good natural ventilation and lighting.

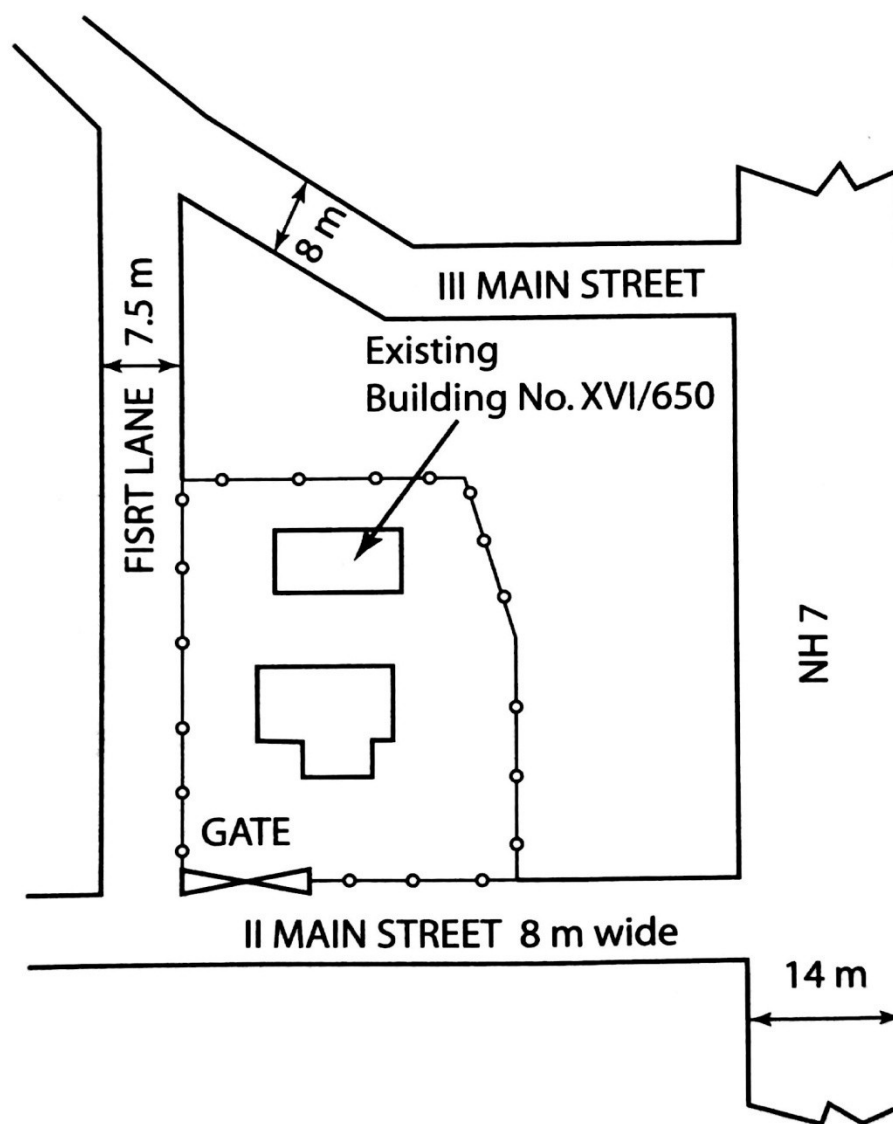
# FACTORS GOVERNING SELECTION OF SITE

- ❑ Site should have quick drainage properties.
- ❑ Flood prone areas, water logged areas and reclaimed land should be avoided.
- ❑ Places prone to air and water pollution should be avoided .
- ❑ Better to have a regular shape with sufficient frontage.
- ❑ Must be sufficient for present and future developments.
- ❑ Type of land use recommended at the proposed site should be checked with local authorities.

# SITE PLAN

- Detailed plan of the plot with necessary surrounding data is called a site plan.
- All the details are to be plotted in the site plan to a scale not less than 1:400 in paper of size 24 cm x 33cm or A3 size.
- While preparing the site plan , prevailing building rules and byelaws are to be taken into consideration.
- In order to obtain sanction for construction, site plan along with key map & service plan is to be submitted to the municipal or town planning authorities.

# SITE PLAN



Survey No.657/5  
Village - Kuzhupilly  
Taluk - Kochi  
District - Ernakulam

Scale 1:400  
Plot area - 455 sq.m  
Plinth area - 125 sq.m

# DETAILS REQUIRED IN A SITE PLAN

- ▣ Boundaries of the plot with the revenue survey particulars
- ▣ Position of the site in relation to the neighbouring streets, with its main access.
- ▣ Details of streets with name, width.
- ▣ All existing structures in the plot.
- ▣ All existing features and footpaths within the plot.
- ▣ Layout of streets adjoining, terminating at the site, existing, proposed to be widened or newly aligned.

# DETAILS REQUIRED IN A SITE PLAN

- ▣ Proposed sub division of plot.
- ▣ The access to each plot sub division.
- ▣ Details of undeveloped, not proposed to be developed and to be reclaimed areas within the plot.
- ▣ North Direction
- ▣ Layout of proposed building.
- ▣ Topographic contours.



# PLANNING REGULATIONS & BYELAWS

- ▢ The building construction activities in an area are monitored by municipal authorities.
- ▢ Building rules and byelaws are framed in a scientific and logical manner.
- ▢ These are formed by a committee of experts comprising of civil engineers, lawyers , traffic officials health officers and general administrative officers.

# KERALA MUNICIPALITY BUILDING RULES,1999

- ▣ Building rules and byelaws are laid down by municipal town planning authorities.
- ▣ The National Building Code,(NBC) published by Government of India acts as a reference for local bodies in framing building bye laws.

# KERALA MUNICIPALITY BUILDING RULES,1999

- Important regulations in Kerala Municipal Building Rules,1999 (KMBR) include
  - i) General requirements regarding plots
  - ii) Exterior & Interior open spaces
  - iii) Built up areas of buildings, coverage and floor area
  - iv) Size, height & ventilation of rooms
  - v) Water supply and sanitary provisions
  - vi) Specifications of parts of the building

# GENERAL REQUIREMENTS REGARDING PLOT

- ❑ Buildings should not be constructed on the land where refuse, excreta and other offensive matter is deposited
- ❑ Plots with big pits, quarries or similar excavations should be avoided unless it is levelled and made suitable.
- ❑ Floodable areas & plots with a slope of more than 45° should not be used for building construction
- ❑ In the areas marked as Coastal Regulation Zone (CRZ) under Environmental Protection Act, construction activities are restricted.

# CLEARANCE FROM OVER HEAD ELECTRIC LINES

| Sl.No | Type of line                 | Vertical clearance        | Horizontal clearance |
|-------|------------------------------|---------------------------|----------------------|
| 1     | Low and medium voltage lines | 2.4 m                     | 1.2 m                |
| 2     | Up to 33 kv lines            | 3.7 m                     | 1.85 m               |
| 3     | More than 33 kv              | 3.7 m plus 0.3m for every |                      |

# EXTERIOR AND INTERIOR SPACES

## □ Exterior Open Spaces

- i) Every building up to 10 m in height shall have a minimum front yard of 3 m depth.
- ii) Every building up to 10 m in height shall have a minimum rear yard of 2 m depth.
- iii) Every building up to 10 m in height shall have an open air space of not less than 1.2 m width on one side and not less than 1.0 m on other sides, other than front and rear yards.

# EXTERIOR AND INTERIOR SPACES

## ■ Exterior Open Spaces

- iv) For buildings above 10 m in height, 0.5m per increase of every 3m height is to be provided apart from minimum front, rear and side open spaces.
- v) For buildings above 4 storeys from ground level , a minimum open space of 5 m is to be provided on any one side contiguous to front side.
- vi) The minimum distance from centre line of street to building should be 4.5 m
- vii) The minimum plot size for residential buildings should be 6m in width and 12 m in depth

# EXTERIOR AND INTERIOR SPACES

## □ Interior Open Spaces

- i) Any habitable room not abutting on front, rear or side open spaces, shall abut an interior open space with a minimum width of 2.4 m
- ii) If the total height of the residential building is less than 7 m, then the interior open space shall have 1.5 m width.



# COVERAGE & FLOOR AREA RATIO

- ▣ Plot Area: Area enclosed between the boundaries of plot.
- ▣ Plinth area: area of the building at the plinth level excluding uncovered staircases, open porch
- ▣ Floor Area: built up area of the building at any floor level. The total floor area means total built up area in all floors excluding parking area.
- ▣ Carpet Area: the usable floor area excluding staircases, lift wells, ducts, toilets, electrical and air condition plant rooms etc.

# COVERAGE & FLOOR AREA RATIO

- ▣ Coverage: The maximum area of any floor of the building excluding cantilevered open balconies, garden, compound wall, gate, uncovered staircases etc.
- ▣ Floor Area Ratio : The ratio of total built up area on all floors to plot area is called Floor Area Ratio(F.A.R)

$$\text{FAR} = \frac{\text{Total Floor Area on all floors}}{\text{Plot Area}}$$

## PROVISIONS OF SIZE, HEIGHT & VENTILATION OF ROOM

- Maximum height of a building is limited to a value equal to two times the width of the abutting road plus two times the front yard. This may be increased at a rate of 3m for every additional 1m setback from building line.
- The minimum headroom in any habitable room shall be 3.0 m. For air conditioned room it shall be 2.4 m.
- The area of habitable room shall not be less than 9.5m<sup>2</sup> with a minimum width of 2.4 m

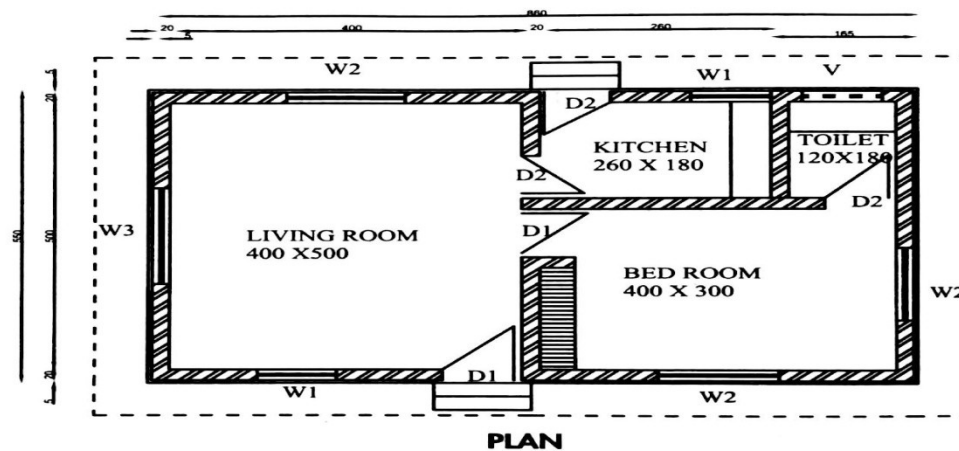
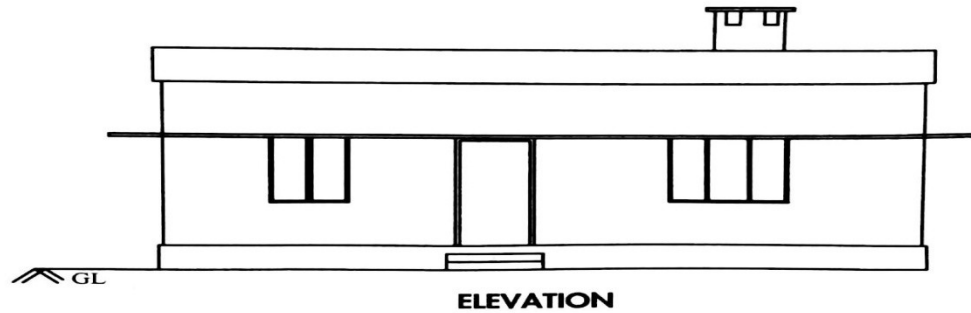
## PROVISIONS OF SIZE, HEIGHT & VENTILATION OF ROOM

- ▣ Every habitable room shall be furnished with sufficient number of openings to facilitate the direct admission of light & air,
- ▣ No portion of the room shall be assumed to be lighted if it is more than 7.5 m away from the opening.
- ▣ Every Kitchen shall be ventilated according to the standards stipulated for habitable rooms and shall have suitable flue for escape of smoke and heated air.
- ▣ The area of bath room shall not be less than 1.5 m<sup>2</sup> with either side not less than 1.1 m,

## PROVISIONS OF SIZE, HEIGHT & VENTILATION OF ROOM

- ❖ Area of the latrine shall not be less than 1.1 sq.m with one side not less than 1.0m
- \* Every bathroom or latrine shall have at least one of its walls abutting on an open air space or open verandas or an air shaft if centrally air conditioned

# A TYPICAL BUILDING PLAN



## INDEX

| MARK | ITEM       | SIZE(cm) |
|------|------------|----------|
| D1   | DOOR       | 90X200   |
| D2   | DOOR       | 80X200   |
| W1   | WINDOW     | 90X120   |
| W2   | WINDOW     | 135X120  |
| W3   | WINDOW     | 180X120  |
| V    | VENTILATOR | 90X60    |

All dimensions are in cm